The architecture of internet aesthetics

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Abstract

Internet visual culture has produced a sprawling vocabulary of "aesthetics" such as Frutiger Aero, vaporwave, cottagecore, and liminal spaces, each a visual genre with a distinct relation to the past. Whether this folk vocabulary has a measurable affective architecture has been argued only qualitatively. Using naturally-occurring human behaviour, we reconstruct the aesthetics ecosystem as a network of 1,115 aesthetics from community cross-referencing. The network is strongly modular, organised by latent affect ($p = 0.0005$) and historical era (1920s to 2010s), and its unsupervised communities align with an a priori four-mode taxonomy (restoration, hauntology, anemoia, uncanny inversion; NMI = 0.50). A second, independent record of user co-curation aligns with the same communities above chance (NMI = 0.21 vs. a 0.17 null, $p = 0.032$). As a sharp test of how real this structure is, a frontier panel (Gemini 2.5 Flash, GPT-4o, Claude Sonnet 4.6) predicts the aesthetic that humans assign each image to from affect ratings alone (accuracy 0.45 vs. 0.29 chance, $z = 18$, $n = 1{,}727$), matching a 29-feature perceptual baseline. The same models over-attribute unease: they reach near-ceiling recall on the uncanny (0.86) while reading warmth and longing as eerie (restorative 0.35, anemoia 0.31). The bias is interpretive: it is absent from image statistics, it persists when the image is edited, and it falls sharply when the rating task is reframed; on standardized affective images the same models match human valence norms ($r = 0.84$ to 0.87). The over-attribution, its boundary, and its causal mechanism replicate across all three models. Internet aesthetics thus have a real, cross-source affective architecture, one legible even to systems never trained on it, which in reading it reveal a diagnosable and correctable bias toward the uncanny. We release the network, corpora, model ratings, and code.

1 Introduction

Nostalgia is usually studied as a single emotion, and the dominant experimental tradition characterizes it as self-relevant, social, and predominantly positive [4, 5, 6, 7], measuring it with instruments that assume the person is retrieving their own lived past. The visual culture that has grown up online behaves very differently. It has produced a sprawling, self-organizing taxonomy of "aesthetics," each a visual-affective genre with its own relation to time, and those relations vary widely.

The four cases below show how widely those relations can differ. Frutiger Aero, the glossy, bubbly, blue-sky-and-grass look of mid-2000s consumer technology, is nostalgia as restored optimism, a remembered future that felt clean and hopeful. Vaporwave takes almost the same era and inverts its affect, recycling corporate muzak and shopping-mall imagery into an ironic elegy for a prosperity that curdled, a nostalgia Fisher calls hauntology, the mourning of "lost futures" [9]. Dark academia and fairycore long for pasts the viewer never inhabited and may never have existed, the case of anemoia that the constructive view of

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memory anticipates [8]. Liminal spaces, dreamcore, and weirdcore keep the ache of nostalgia while draining its comfort, producing imagery that is familiar and wrong at once, nostalgia inverted into the uncanny.

Two assumptions of the standard account are strained by these cases. The first is that nostalgia rests on autobiographical memory: much of the audience feeling nostalgia for Frutiger Aero were children or unborn during its run, a clean instance of nostalgia for an unlived time. The second is that nostalgia is positively valenced: the inverted aesthetics are unmistakably nostalgic and unmistakably unsettling. The uncanny here belongs to the same family as nostalgia, an inversion of it.

These aesthetics raise two questions that have been treated only qualitatively. The first is whether this folk vocabulary has a hidden affective structure that can be recovered, without a rating study, from the traces of how people collectively sort and relate aesthetics online. The second, a sharp test of how real that structure is, asks whether today's frontier vision-language models, trained on the same web, perceive it, and where their reading of culturally constructed affect breaks down. We answer the first from two large behavioural records: the community-built Aesthetics Wiki, whose editors link each aesthetic to "related" ones, and the curation platform are.na, where users file images into named channels. Both are naturally-occurring records of how a visual culture organizes itself. We answer the second with a three-model frontier panel (Gemini 2.5 Flash, GPT-4o, Claude Sonnet 4.6), evaluated against those human records and against standardized human affect norms. Our contributions:

1. Internet aesthetics have a recoverable, modular, cross-source affective architecture. A network of 1,115 aesthetics built from collective curation is strongly modular ($Q = 0.53$) and organized by latent affect and historical era, and a second, independent record of user co-curation aligns modestly but significantly with the same communities.

2. Its unsupervised communities align with an a priori four-mode taxonomy (restorative, hauntological, anemoia, inverted; NMI = 0.50), so the sprawling folk vocabulary maps onto a small set of affective operations on the past.

3. Frontier vision-language models read this structure and over-attribute unease. A frontier panel predicts, from affect ratings alone, the aesthetic that humans assign an image to, as accurately as a 29-feature perceptual description ($z = 18$); the same panel reads warm and wistful genres as eerie (near-ceiling recall on the uncanny, near chance on restorative and anemoia). The bias is interpretive (absent from image statistics, unchanged by editing the image, reduced by reframing the rating task) and bounded (the models match human valence norms on standardized images, $r = 0.85$).

4. Open resources: the aesthetics network with an interactive explorer, an openly-licensed image corpus and feature table, the three-model affect ratings, and all harvest and analysis code.

We collect no first-party ratings; every affective criterion is an existing human record or a standardized human norm, which keeps the analysis naturalistic and at scale.

Computational affect from images is well established: large emotion datasets pair images with human labels [25, 27, 28, 29], and subjective image properties such as memorability are predicted to near human consistency by deep features [26]. Vision-language models are increasingly evaluated as affect raters [30]; their reliability is uneven and weakest for higher-order, culturally specific interpretation [31], with documented intensity miscalibration on affective content [2]. The language-model literature reports systematic, sometimes human-absent biases [33, 34] and exaggerated-safety behaviour, in which models treat benign inputs that merely look risky as unsafe [32]. We use VLMs here as a measurement instrument turned on a culturally constructed stimulus space, and identify a directional affective bias, an over-attribution of unease to stylized genres, that is the visual analogue of exaggerated safety and that operates at the level of interpretation.

2 Methods

We group aesthetics by their affective operation on the past; the folk labels are many and overlapping, while the operations are few. Table 1 lists the mechanism clusters and Fig. 1 shows representative stimuli.

Table 1: Corpus (N = 681), by affective-mechanism cluster. All images openly licensed (Wikimedia Commons). Named aesthetics are folded into the cluster by their operation on the past; retro-futurism is split across restorative (optimistic) and hauntological (decayed) strands.

cluster	representative named aesthetics	n
restorative	Frutiger Aero, Y2K, cottagecore, synthwave, retro-futurism (optimistic)	160
hauntological	vaporwave, mallsoft, cassette-/diesel-futurism (decayed retro-futurism)	123
inverted	liminal spaces, the Backrooms, poolrooms, dreamcore, weirdcore	194
anemoia	dark academia, fairycore, royaltycore, old money	84
control	ordinary populated present-day scenes (non-nostalgic baseline)	120
total		681

Restorative (R). Warm, comfort-seeking, techno-optimist: Frutiger Aero, Frutiger Eco, Y2K/Cybercore, cottagecore, kidcore, and the optimistic strand of retro-futurism (Googie, atomic-age, synthwave). The past, or the past's imagined future, recalled as good (Boym's restorative pole; 1).

Hauntological (H). Ironic, melancholic, ambivalent: vaporwave, mallsoft, chillwave, and the decayed strand of retro-futurism (cassette futurism, dieselpunk). Fisher's lost futures: longing inflected by the knowledge that the promised future did not arrive.

Anemoia (A). Longing for unlived, often never-existent pasts: dark academia, light academia, fairycore, goblincore, royaltycore, old money. Nostalgia fully detached from autobiography.

Inverted / uncanny (I). Nostalgia stripped of comfort: liminal spaces (the term adapted from ritual liminality, the unstable in-between phase of a rite of passage; 11), the Backrooms, poolrooms, dreamcore, weirdcore, traumacore, analog horror. The eerie in Fisher's sense, where something that should be present is absent, as in a place with no people [10]. This family connects nostalgia to the uncanny proper [12, 13, 14], whose modern accounts emphasize realism inconsistency [15] and configural deviation from the familiar [16, 17].

Control (C). Ordinary, populated, present-day scenes: the non-nostalgic baseline against which the others are measured, and the reference distribution for the ϕ coordinates.

Each image is reduced to a compact set of image-computable coordinates: three perceptual axes and four mechanism coordinates ϕ. The perceptual axes capture colour and gloss (the restorative signature), structural emptiness and human absence from ADE20K segmentation [22] (the eerie signature), and an expected-but-absent object score from open-vocabulary detection [23]. The four ϕ coordinates place each image relative to ordinary present-day scenes: typicality (distance from the per-scene control centroid), surprise (reconstruction error off the control manifold), realism variance, and scene entropy. The reference for every ϕ is estimated on present-day controls alone, which keeps the coordinates independent of any model rating or interpretation applied later. Full definitions, the DINOv2/CLIP backbone, and statistical procedures appear in Supplementary Material S1-S4. Together the axes and ϕ form the affective-temporal space, and the clusters are regions within it.

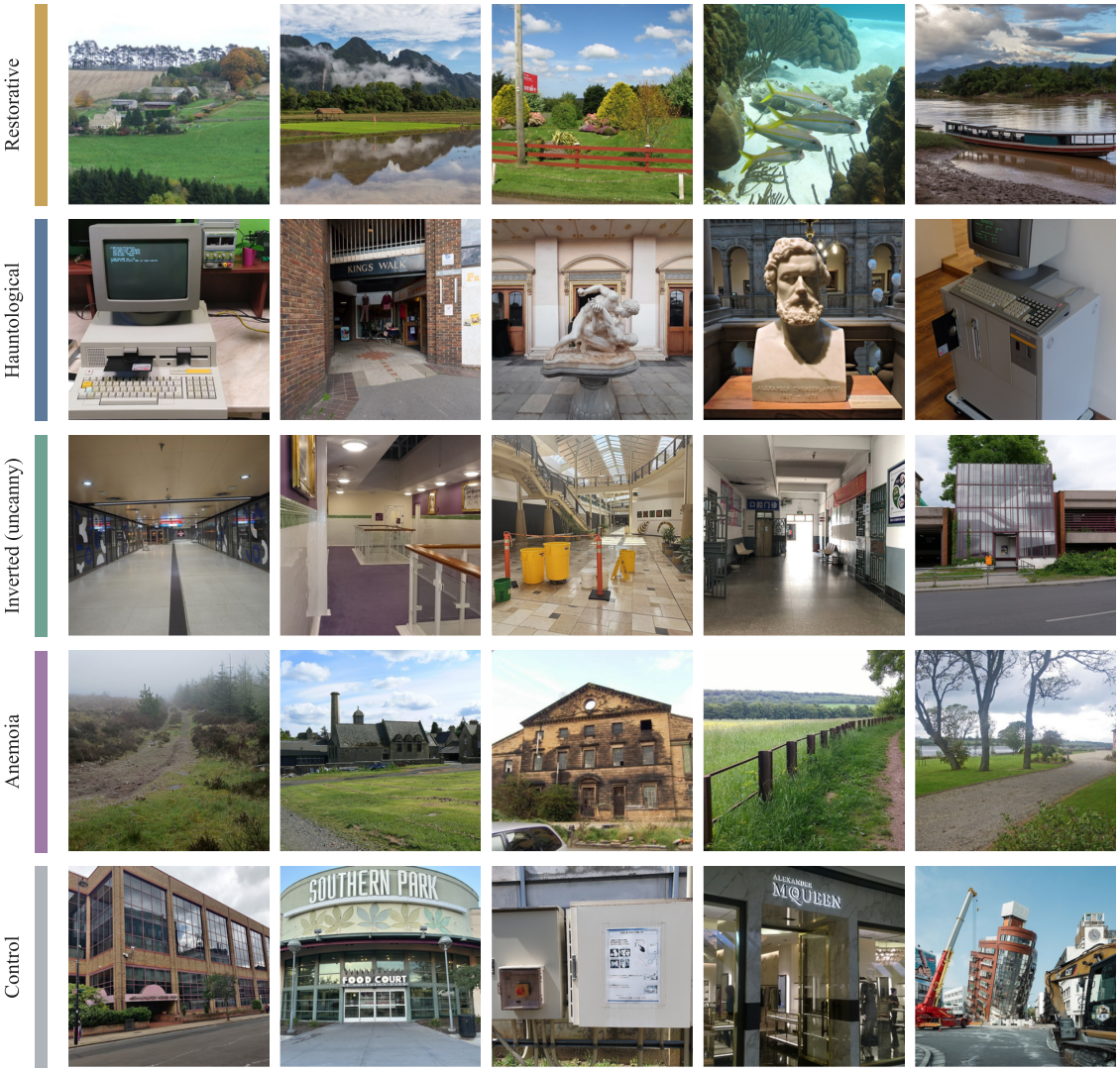


Figure 1: Representative stimuli per cluster (the five images nearest each cluster centroid in standardized feature space; genuine exemplars). Top to bottom: restorative (nature/water/sky, glossy optimism), hauntological (busts, CRT-era electronics: "lost futures"), inverted (empty corridors, malls, lots: liminal/kenopsia), anemoia (estates, misty forests, libraries), control (the same scene types as inverted, populated).

3 Results

The evidence arrives in three movements: a global map of how the aesthetics relate, a check that the map survives when rebuilt from a second and unrelated record of behaviour, and a test of whether the same structure is legible in the images themselves and to frontier vision-language models.

The map comes first. Treating the Aesthetics Wiki as a behavioural record (Methods; Supplement S6), we enumerate every article that carries an aesthetic infobox and parse its RELATED_AESTHETICS field, which yields a network of 1,115 aesthetics joined by 5,889 human-drawn links. It forms a single connected component, so internet aesthetics are one cultural fabric rather than a scatter of isolated scenes, and that fabric is highly structured: Louvain community detection finds 13 communities with modularity $Q = 0.526$, against a degree-preserving rewired null of $Q = 0.007$ (Fig. 2).

Two latent axes organise that partition, and the first is affect. Scoring each aesthetic's community-authored KEY_VALUES on a fixed four-dimension lexicon (unease, nostalgia, comfort, refinement), edges that join two aesthetics of the same dominant affect outnumber a label-permutation null by a wide margin (537 of 1,249 such edges, against 353 expected; $p = 0.0005$). Affect organises the graph, and it does so across the grain: the communities are mostly subcultural (goth, punk, steampunk; cottagecore and nature; academia; futurism), with affect running as a gradient within and between them rather than carving out communities of its own. The second axis is time. Community median DECADE_OF_ORIGIN climbs from the 1920s (modernism, art nouveau) through the 1950s (mid-century, retro-futurism) and 1980s (punk, goth) to the 2010s (the internet-native cluster), a second and orthogonal organiser.

The four-way affective-mechanism taxonomy we specify on theoretical grounds (restorative, hauntological, anemoia, inverted) aligns with the network's unsupervised communities: locating our 23 seed aesthetics in the network, the normalised mutual information between their a-priori cluster and their (label-free) community is NMI = 0.50 (permutation null 0.27, $p = 0.002$). The alignment is partial, with restorative aesthetics in particular scattering across several communities, and it sits well above chance, obtained with no researcher labels entering the community detection.

The map so far rests on a single behavioural trace, the editors' RELATED links, so its modularity could in principle reflect how one community writes cross-references. The second movement rebuilds the network from unrelated behaviour: are.na user co-curation. For every harvested image we retrieved the full set of channels users filed it into, mapped channel titles to aesthetics, and linked two aesthetics whenever users co-filed the same image under both, a record of curation behaviour that shares no mechanism with wiki editing. The resulting co-curation graph (104 aesthetics, 354 edges) has sensible structure (top pairs: Fairycore and Fairytale, Cassette Futurism and Futurism, Dreamcore and Weirdcore), and its communities align with the wiki communities above chance on the 103 shared aesthetics (NMI = 0.21 versus a label-permutation null of 0.17, $p = 0.032$). The agreement is modest and significant: two independent traces of collective behaviour converge, which places the community structure beyond a single editorial source.

The third movement turns from how people relate aesthetics to the images themselves, where the modes are visible directly in content. On the openly-licensed corpus ($N = 681$; Wikimedia Commons), a five-way classifier over the perceptual features separates the modes above chance (cross-validated accuracy 0.57; permutation $p < 0.01$), and the inverted/uncanny pole occupies a region of high structural human-absence, separating from scene-matched populated controls within category (Cohen's $d = 0.38$, 95% CI $[0.14, 0.68]$). An independent CLIP observer reproduces the same inverted-vs-rest eerie ordering ($d = 0.72$, 95% CI $[0.47, 0.99]$), foreshadowing the frontier-model analysis below. Feature definitions, the classifier decomposition, the per-scene cells, and the CLIP probes are given in Supplementary Sec. S7.

CLIP encodes human associations only indirectly, so we tested the taxonomy against two sources of explicit collective human labelling, harvested from the community-run Aesthetics Wiki and the curation platform are.na (Supplement S6). Both records are observational, and we state their limits below; each provides a human criterion external to our features and our corpus.

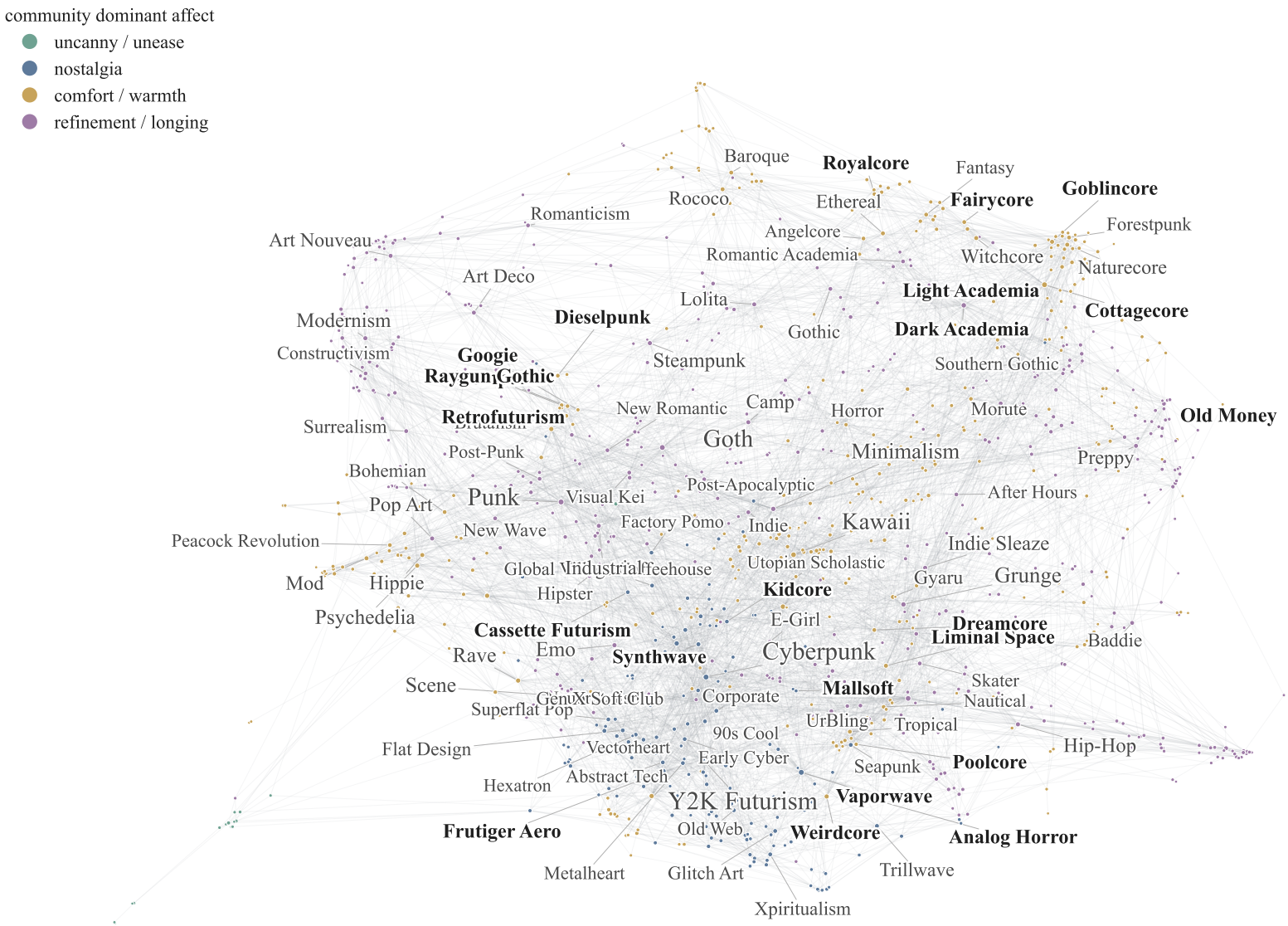


Figure 2: The internet-aesthetics curation network. 1,115 aesthetics joined by 5,889 community-authored "related" links, laid out by a force-directed algorithm (ForceAtlas2). Nodes are coloured by their community's dominant community-authored affect and sized by curatorial degree. All 1,115 aesthetics are plotted; 100 are labelled (the 26 focal aesthetics in bold, the highest-degree hubs in grey, font scaled by degree); an interactive explorer of the full set is released with the data. The network is strongly modular (13 Louvain communities, $Q = 0.53$ vs. a 0.01 rewired null), and communities separate systematically by affect ($p = 0.0005$) and by era (1920s to 2010s).

The first source is community-authored affect: each Aesthetics-Wiki page carries a structured KEY_VALUES field naming the aesthetic's affective content, and 21 of our 26 named aesthetics populate it. Scoring these against a pre-specified four-dimension affect lexicon (unease, nostalgia, comfort, refinement) recovers the predicted structure. The uncanny (inverted) cluster dominates unease ($d = 2.66$ versus all other clusters pooled); restorative aesthetics are described overwhelmingly in terms of comfort, and anemoia almost entirely in terms of refinement (intellectualism, elegance, tradition). Nostalgia is named across restorative (50% of aesthetics), hauntological (75%), and inverted (80%) aesthetics, direct community evidence that nostalgia is a shared substrate across modes, the paper's central claim. Anemoia is the informative exception: the community labels it with refinement terms and uses the word nostalgia for none of its aesthetics (0%), consistent with anemoia being a longing for an unlived past that feels distinct from remembered nostalgia.

The second source is a crowd-drawn similarity graph: each page also lists RELATED AESTHETICS, yielding a crowd-authored similarity network. Restricting to the 73 edges whose endpoints both fall in our taxonomy, the fraction joining two aesthetics of the same affective-mechanism cluster is 0.49, against a label-permutation null of 0.23 ($p = 0.0006$, 5000 permutations). Independent human judgements of which aesthetics "go together" thus recover the R/H/I/A grouping far above chance, and unlike the lexicon analysis this test uses no researcher-defined word list, so it avoids circularity.

The two preceding tests are category-level; the strongest test is whether our image features predict, at the image level, the aesthetic that humans assigned each image to. We harvested ≈1,900 are.na images whose channel placement is a crowd membership label, extracted the same feature set, and asked a random forest to predict the label under channel-grouped cross-validation, holding all images from one channel out together so the model cannot exploit a single curator's house style or near-duplicates (Supplement S6). Structural features predict the four-way mechanism cluster at 0.42 (majority baseline 0.29; permutation null 0.28; $z = 12.4$, $p < 0.005$, 200 shuffles) and the 20-way specific aesthetic at 0.13 (chance 0.07; $z = 10.2$, $p < 0.005$): well above chance, and modest in absolute terms, as the heavy cluster overlap predicts. Two facts echo the main analysis: dropping the ϕ coordinates barely changes accuracy (0.41), so the perceptual axes carry the signal; and anemoia is again the worst-recovered class, dispersing into the three other modes, which marks it as the least distinct visual mode. The confusions track genuine visual kinship (Frutiger Aero with Y2K, vaporwave with cassette futurism, dark academia with old money).

The generalizable signal is in collective association. Training the classifier on one corpus and predicting the other, disjoint corpus (are.na, human-curated; Wikimedia, keyword-sourced) transfers above chance (are.na→Wikimedia accuracy 0.38 vs. 0.35 majority, $z = 3.2$; reverse 0.32 vs. 0.30, $z = 2.5$). The structure that generalizes across independent records therefore resides chiefly in how people collectively curate and relate aesthetics, consistent with the network and co-curation results above, and less in low-level image statistics specific to any one source.

Three tests across two unrelated sources of collective human labelling (category affect terms, a crowd similarity graph, and image-level membership) corroborate a taxonomy derived on theoretical grounds, with no rating study run.

The collective-human structure invites a machine-cognition question that needs no rating study: whether frontier vision-language models (VLMs), trained on web-scale human image-text data and never on this taxonomy, perceive the same affective structure, and where their reading fails. We assembled a three-model panel (Gemini 2.5 Flash, GPT-4o, and Claude Sonnet 4.6; API identifiers `gemini-2.5-flash`, `gpt-4o`, `claude-sonnet-4-6`) and had each rate every are.na image on calibrated 1-5 affect scales (eerie, nostalgic, abandoned, dreamlike, valence, arousal) and force-choose one mechanism mode. We used rubric anchors and a "reserve 5 for unambiguous cases" instruction because VLM affect ratings are known to inflate, and a three-judge consensus is the recommended mitigation [2].

The judges agree, and they track human curation. Inter-model agreement is high (pairwise Pearson on the 1,115 images rated by all three: eerie $r = 0.76$, dreamlike 0.77, nostalgic 0.74, valence 0.72, arousal 0.70; mode agreement 0.70), so the consensus is a reliable, shared measurement. The six consensus affect

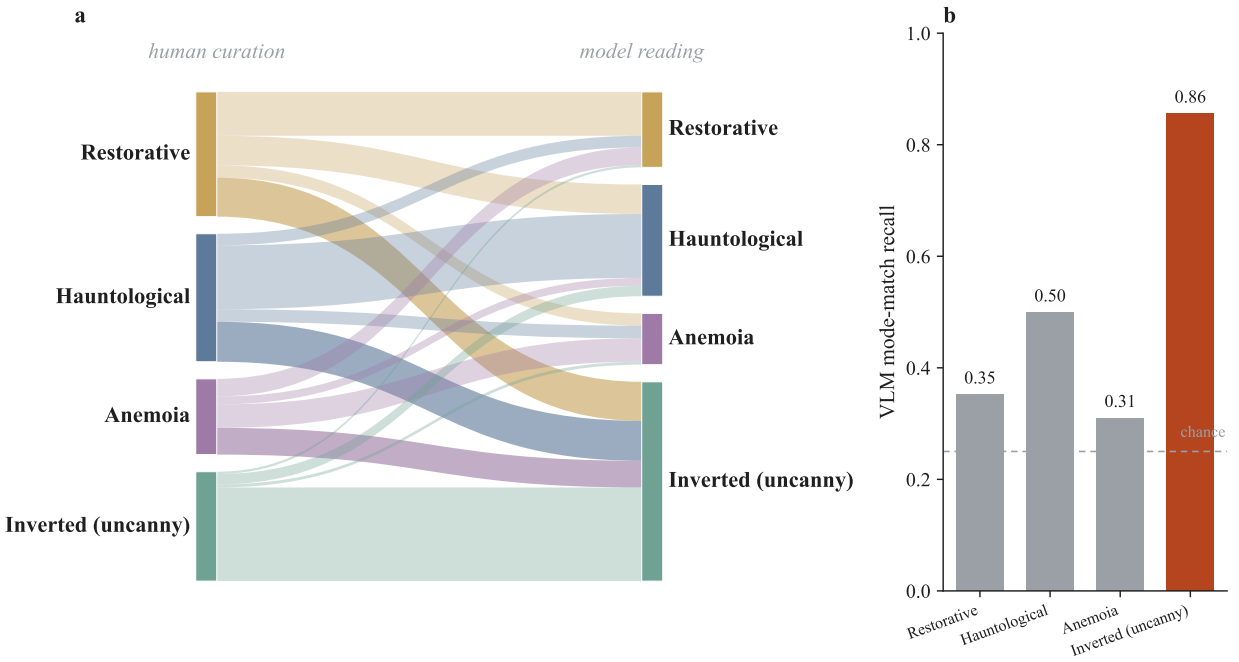


Figure 3: The asymmetric leak into the uncanny. Three-model consensus (Gemini 2.5 Flash, GPT-4o, Claude Sonnet 4.6) over 1,885 are.na images. (a) Flow from human-assigned mode to VLM consensus mode: each warm genre sends about a third of its images into "inverted" (restorative 0.31, hauntological 0.31, anemoia 0.35), while "inverted" retains 0.86 of its own and leaks little back, so unease acts as a sink. (b) Per-mode mode-match recall: near-ceiling for the uncanny and near chance for the warm genres. The full confusion matrix appears in the Supplement (Fig. S4).

axes predict the aesthetic that humans assigned each image to: under channel-grouped cross-validation, a random forest on the affect axes of the two fully-covered judges reaches 0.45 accuracy across the four modes (majority 0.29; permutation null 0.26 ± 0.01; $z = 18.1$, $n = 1{,}727$). Adding the third judge on the images it covers leaves this unchanged (two-judge 0.60 vs. three-judge 0.62 on that subset, where the majority baseline is itself 0.48), so the prediction is robust to model composition. It matches the 0.42 achieved by our hand-built structural features, at a higher effect size: a handful of numbers a frontier model emits about an image's feeling recover human curatorial categories as well as a 29-dimensional perceptual description.

The models' categorical judgements expose a systematic, directional failure (Fig. 3). Against the human modes, consensus mode-match is near-ceiling for the inverted/uncanny class (recall 0.86) and much lower for restorative (0.35) and anemoia (0.31). The misclassifications are systematic: warmth and longing leak specifically into "inverted" (restorative→inverted 0.31; anemoia→inverted 0.35), and all three models independently label the techno-optimist flagship Frutiger Aero as "hauntological." Frontier VLMs over-attribute unease, reading the eerie into images humans experience as comforting or wistful. The effect is affect-inflation localized to a single, culturally loaded axis, the uncanny, and it is the paper's clearest machine-cognition result: for these models the difficult operation is to withhold the uncanny.

A human-anchored boundary test separates two possibilities: the over-attribution could be a generic VLM negativity bias, or something specific to internet aesthetics. We tested it against human ground truth on OASIS [3], ~890 images with normative valence/arousal ratings ($N \approx 100$ each), rated by all three models. Every model tracks the human valence norm closely (Pearson $r = 0.85, 0.84, 0.87$ for Gemini, GPT-4o, Claude), with a small negativity offset (-0.02 to -0.07 on a 0 to 1 scale), and all three reserve low EERIE for human-positive images (mean 1.2 to 1.3 of 5). On ordinary affective images the models are well-calibrated against people. The over-attribution is therefore domain-specific to internet aesthetics: the failure is localised

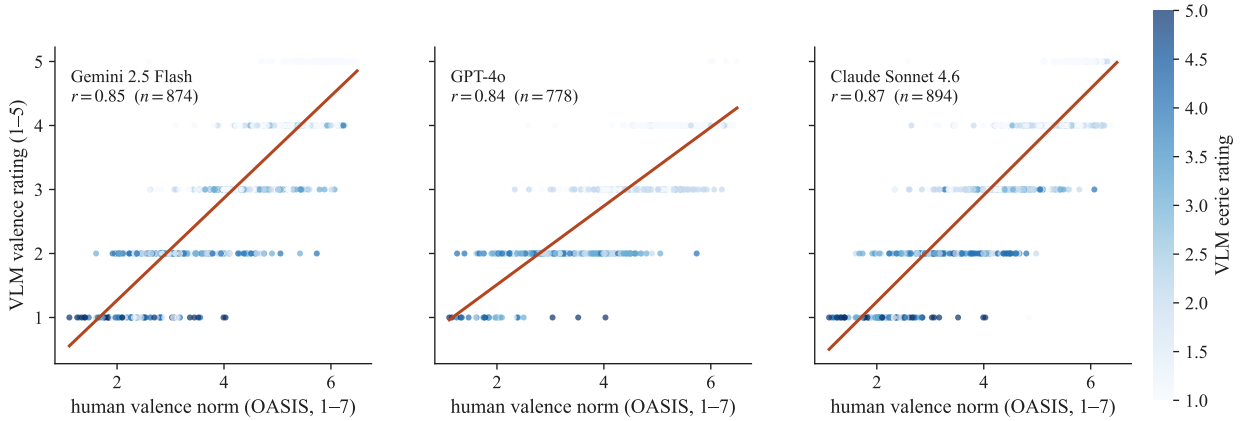


Figure 4: The bias is bounded: models are calibrated on standard affective images. VLM valence vs. human-norm valence on OASIS, per model; points coloured by the model's EERIE rating. All three track human valence ($r = 0.85/0.84/0.87$) and concentrate high eerie (dark) at low human valence, reserving unease for genuinely negative scenes. On standard affective images the models stay calibrated; the over-attribution appears on internet aesthetics.

to the ambiguous, culturally-constructed visual genres, which makes it a finding about cultural perception (Fig. 4).

We traced the cause by dissecting the are.na images humans filed as restorative or anemoia while a model rated them eerie (≥ 3), with one correlational and two causal tests.

Within these naturally-warm images, a correlational test shows the model's EERIE rating is barely predicted by low-level image statistics: brightness, saturation, colourfulness, edge density, gloss, blur (Laplacian variance) and resolution together explain almost none of the variance (standardised multiple regression $R^2 = 0.04$; the only sizeable term is structural emptiness, $\beta = 0.22$). Eeriness stands independent of degradation and palette.

A first intervention edits the image: we naturalised 120 such images by denoising, upscaling and sharpening (quality), and by restoring brightness, contrast and saturation (tone), then re-rated them. Every manipulation left eeriness at or above its original level (full naturalization: Gemini $\Delta = +0.08$, 95% CI $[-0.04, +0.21]$ and GPT-4o $+0.06$ $[-0.03, +0.15]$, both n.s.; Claude $+0.12$ $[+0.03, +0.22]$). Making an internet-aesthetic image look like a clean photograph leaves the attributed unease in place, so its cause lies beyond the processed appearance.

A second intervention changes only the question, holding the image fixed. Reframing the rating to "the scene depicted, rather than its visual style" lowered eeriness in every model, and an explicit debias ("stylisation, such as filters, grain, low resolution, text, and retro grading, is not evidence of threat") lowered it further: $\Delta = -0.72$ (95% CI $[-0.87, -0.58]$) for Gemini, -0.24 $[-0.33, -0.16]$ for GPT-4o, and -0.37 $[-0.50, -0.23]$ for Claude, all reliably below zero. The pixels were fixed, and a one-sentence reframe moved the rating toward calibration in all three (Fig. 5).

The convergent picture is that frontier VLMs treat stylisation itself, the visual signature of a curated internet aesthetic, as evidence of wrongness, and they do so at the level of interpretation: the bias is invisible in the image statistics, it survives editing of the image, and it is largely correctable by reframing the task. This is the visual, affective counterpart of exaggerated safety in language models, where benign inputs that merely look risky are treated as unsafe [32], and of the broader tendency of frontier models to amplify, and sometimes invent, evaluative biases [34, 33]; here the surface cue is a genre's post-processing in place of a keyword. It also yields a practical fix: affective-computing pipelines that elicit scene-level affect, in place of style-level affect, substantially debias VLM unease.

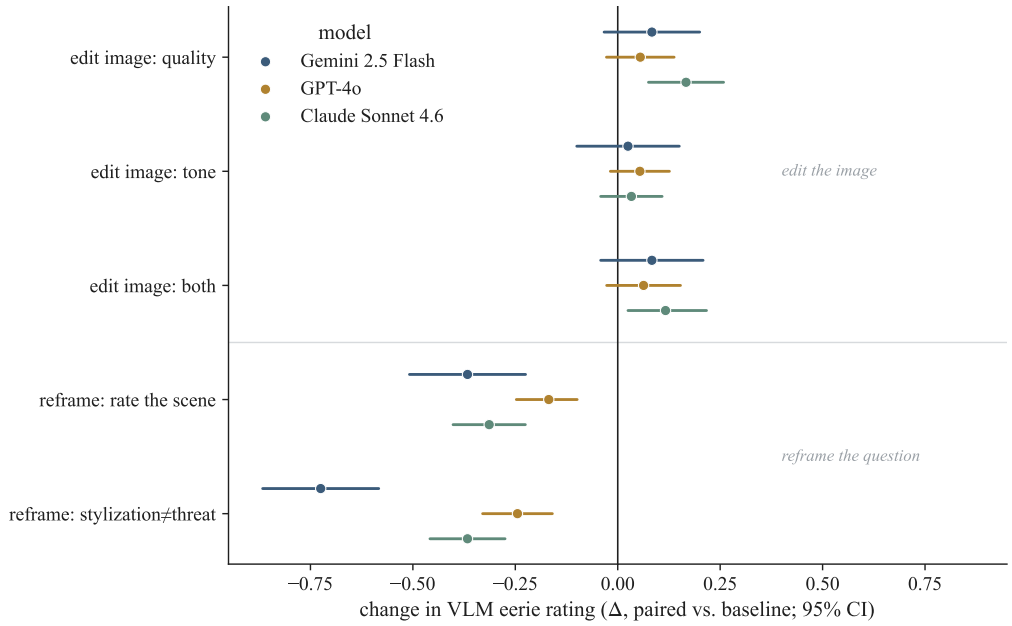


Figure 5: The over-attribution operates at the level of interpretation. Paired change in VLM EERIE for each manipulation, per model (95% CI). Editing the image (denoise, upscale, recolour) never lowers eeriness (intervals at or above zero) in any model; reframing the question (rate the scene; treat stylisation as benign) lowers it (intervals left of zero). The cause lies in interpretation.

4 Discussion

Three independent windows on internet visual culture (how an online community cross-references aesthetics, how users co-curate images, and how frontier vision-language models perceive those images) converge on a common affective architecture, and that convergence is this paper's throughline. The architecture stays consistent across independent records: a network built from editors' cross-references is strongly modular and organized by latent affect and historical era; a second record of user co-curation aligns with the same communities above chance (NMI = 0.21 vs. a 0.17 null, $p = 0.032$); and those unsupervised communities align with the four-mode taxonomy (NMI = 0.50). The signal is carried chiefly by collective association, how people group and curate aesthetics, and only in part by low-level pixels, which transfer across sources weakly ($z \approx 3$). What the analysis recovers is the culture's own affective organization, read from behaviour.

The architecture's most striking property is that systems never trained on it perceive it: a frontier panel predicts human curatorial categories from affect ratings alone, as accurately as a 29-dimensional perceptual description ($z = 18$). That legibility is biased in a specific direction: the models reach near-ceiling on the uncanny while reading warmth and longing as eerie, over-attributing unease along the most culturally loaded axis. Three diagnostics localize the cause: the over-attribution is absent from image statistics, it is unchanged by naturalizing the image, and it falls sharply when the rating question is reframed, so it operates at the level of interpretation, with the models treating stylization itself as evidence of wrongness. Across the panel's affect space the eerie field forms a shadow that spreads from the uncanny pole into the warm territories, the same over-attribution seen image by image (Fig. 6). It is also bounded: on standardized affective images the models match human valence norms ($r = 0.84$ to 0.87) and reserve eeriness for genuinely negative scenes. The phenomenon, its boundary, and its causal mechanism hold across Gemini, GPT-4o and Claude, so the bias is a property of current frontier VLMs broadly. It extends the exaggerated-safety phenomenon documented for language inputs [32] into culturally coded visual affect: a directional, diagnosable, bounded,

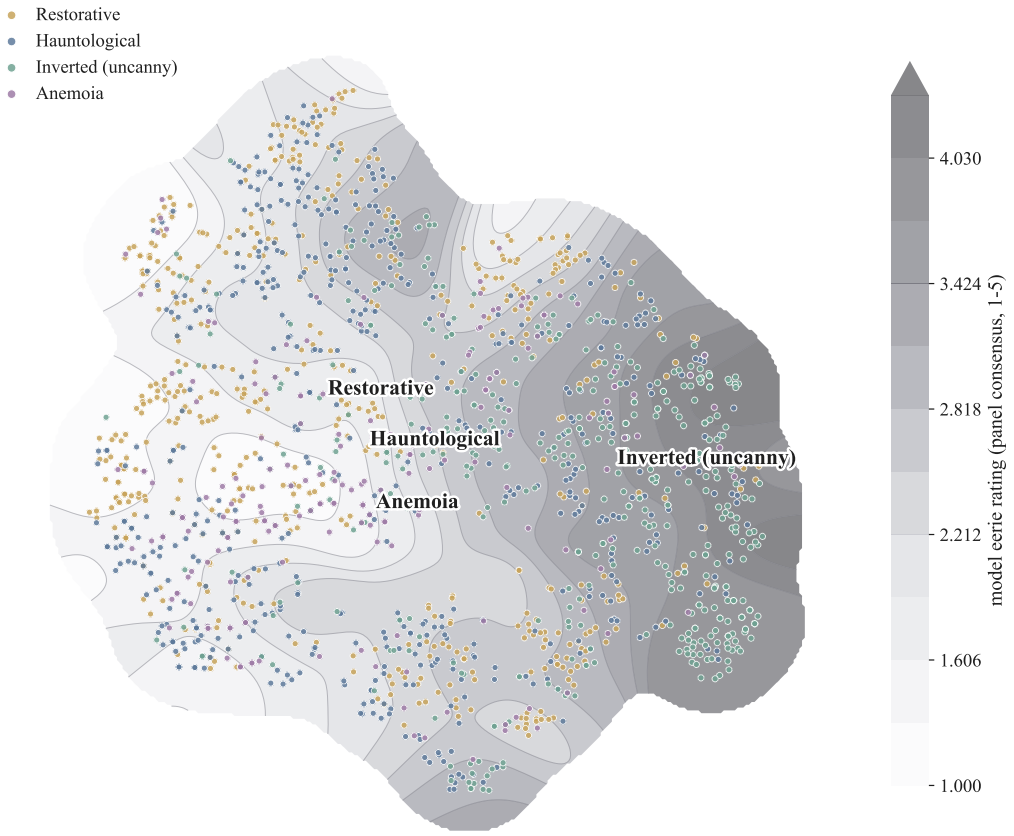


Figure 6: The panel's affective landscape. Each are.na image is embedded (t-SNE) by the panel's non-eerie affect ratings (valence, arousal, nostalgic, abandoned, dreamlike) and coloured by its human-assigned mode; the eerie rating, withheld from the embedding, is overlaid as a smoothed shadow field. Eerie rises toward the uncanny pole, and warm-labelled images carry a modest eerie elevation throughout. The map is a descriptive view of the model's affect space; the causal evidence for the over-attribution is Figs. 3–5.

and correctable bias, with direct consequences for affective-computing systems that elicit style-level affect in place of scene-level affect.

These results also bear on nostalgia itself, since the aesthetics of the feed externalize at least four operations on the past, restoring it, mourning its lost futures, inventing one never lived, and inverting it into unease, and both the human records and the models place the uncanny within nostalgia, as its inversion. Anemoia is the least pictorially distinct of the four, dispersing across communities and proving the hardest mode for both features and models, consistent with its coherence being conceptual, a shared relation to an unlived past, as much as visual.

5 Limitations

Three boundaries frame the scope of these results. First, the evidence is observational: the affective criteria are existing human records (community descriptions, are.na curation) and standardized norms in place of first-party ratings, which buys naturalism and scale while ceding experimental control over what each criterion isolates. Second, the are.na curators and wiki editors are a design-literate online population, so

the recovered "human" structure reflects how that community organizes visual culture. Third, the openly-licensed image corpus indexes the subject matter of some aesthetics more than their post-processing, which is why the central image-level and model analyses use the are.na corpus, curated by people as instances of each aesthetic. The over-attribution, the OASIS boundary, and the causal mechanism are each established across all three frontier models (Gemini, GPT-4o, Claude); we cannot say whether the effect extends to closed reasoning models or to future architectures.

6 Conclusion

Internet visual culture has a measurable affective architecture, and that architecture is legible to the machines now trained on the same culture. Reading two independent records of collective curation, we recover a modular network of aesthetics organized by affect and era, and a four-mode taxonomy that its unsupervised communities align with. A frontier panel recovers the same structure from images alone, predicting human curatorial categories from affect, and it over-attributes unease to culturally constructed warm and wistful genres. We trace that bias to interpretation: it is absent from the pixels, it survives editing of them, it is bounded (the models match human affect norms on standardized images), and it falls substantially when the rating task is reframed. The result is a diagnosable, bounded, and correctable failure mode of frontier affective perception, with a practical fix for systems that read affect from culturally stylized imagery.

Data and code availability. We release, as an open archive on Zenodo (DOI: 10.5281/zenodo.20698074): the aesthetics catalogue and ecosystem network (1,115 aesthetics with affect, era, and related-aesthetic links), with a self-contained interactive HTML explorer of the full network (pan, zoom, search); the are.na human-label and co-curation records; the openly-licensed 681-image corpus with per-image attribution and its feature table; the three-model frontier affect ratings and the OASIS benchmarking and mechanistic-intervention outputs; and all harvest and analysis code. The compilation is CC-BY-4.0; licensed images retain their individual CC / public-domain licences. are.na images are copyright-held and were used for feature extraction only and are not redistributed.

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Supplementary Material

The main text reports what each quantity measures; this supplement gives the formal definitions, the model backbone and implementation, the statistical procedures, and the computational-observer probes, so the pipeline is fully specified and reproducible.

S1 Perceptual feature definitions

Colour and gloss: on an image in linear RGB we compute a CIELAB k-means palette, saturation-weighted circular hue statistics, and the Hasler-Süsstrunk colourfulness

$$C = \sqrt{\sigma_{rg}^2 + \sigma_{yb}^2} + 0.3\sqrt{\mu_{rg}^2 + \mu_{yb}^2}, \qquad rg = R - G, \;\; yb = \tfrac{1}{2}(R + G) - B. \tag{S1}$$

The gloss index is $g = \rho_{\text{smooth}}/(\rho_{\text{edge}} + \varepsilon)$, where ρ_{smooth} is the fraction of pixels with a gentle, non-flat, non-edge luminance gradient and ρ_{edge} is the Canny edge fraction.

Structural emptiness: with structural-surface, occupant, and person area fractions $f_{\text{struct}}, f_{\text{occ}}, f_{\text{person}}$ from ADE20K segmentation,

$$\textsc{emptiness} = f_{\text{struct}}\big(1 - \min(1,\, 3 f_{\text{occ}})\big), \tag{S2}$$

$$\textsc{human_absence} = f_{\text{struct}}\big(1 - \min(1,\, 50 f_{\text{person}})\big). \tag{S3}$$

The factor 50 makes HUMAN_ABSENCE collapse as soon as any person area appears, isolating absence as distinct from bareness. We also compute a vanishing-structure measure (longest uninterrupted structural run) and region-count / size-Gini measures.

Object expectation: with an open-vocabulary detector and a per-scene object prior $p(o \mid s)$,

$$\textsc{exp_absent}(s) = \frac{1}{Z}\sum_o p(o \mid s)\,\big(1 - d(o)\big), \qquad Z = \sum_o p(o \mid s), \tag{S4}$$

with $d(o) \in [0, 1]$ the detector's maximum confidence for object o.

S2 Mechanism coordinates ϕ

Let z be the DINOv2 [CLS] embedding. The reference distribution is estimated only on present-day control images, grouped by scene category c (the firewall).

$$\phi_{\text{typ}}(z, c) = -\,(z - \mu_c)^\top \Sigma_c^{-1}\,(z - \mu_c) \qquad \text{(negative squared Mahalanobis; } \Sigma_c^{-1} \text{ Ledoit-Wolf shrinkage)} \tag{S5}$$

$$\phi_{\text{surp}}(z) = s(z) - \hat{\beta}^\top c_{\text{lo}} \qquad s(z) = \|z - (\mu + WW^\top(z - \mu))\|^2,\ W \text{ a control-fit PCA basis} \tag{S6}$$

$$\phi_{\text{real}} = \text{Var}_t\; r_t, \;\; r_t = P(\text{real photo} \mid \text{tile } t) \qquad \text{(CLIP, } 4 \times 4 \text{ tiles)} \tag{S7}$$

$$\phi_{\text{ent}}(z) = -\textstyle\sum_k p_k \log p_k \qquad (p = \text{softmax of a control-trained scene classifier}). \tag{S8}$$

In ϕ_{surp} the raw reconstruction error $s(z)$ is residualized on low-level complexity covariates c_{lo} (JPEG compressibility, edge density, radial spatial-frequency slope) by a regression $\hat{\beta}$ fit on controls only, so the coordinate reflects semantic surprise with textural surprise regressed out. ϕ_{surp} is a pragmatic stand-in for a trained generative model's reconstruction NLL.

S3 Implementation

SegFormer-B4 (ADE20K) for segmentation, OWLv2-base-ensemble for open-vocabulary detection, DINOv2-base for embeddings, and CLIP ViT-B/32 for realism variance and the computational-observer probes (S5). All models load lazily. The pipeline runs on an NVIDIA RTX 5070 Ti (Blackwell, sm_120), which requires a CUDA-12.8 PyTorch build (the default wheel fails on this architecture); a full 681-image run completes in $\approx$ 11 minutes.

S4 Statistical procedures

Cluster separability used a five-way classifier (random forest, 400 trees; and multinomial logistic) under stratified 5-fold cross-validation, with a permutation baseline of 200 label shuffles. The firewall decomposition re-ran the random forest on the 26 non-ϕ features (13 colour, 10 segmentation, 3 object; EXP_ABSENT excluded as scene-tagged only) and on the three all-image ϕ features. Scene-matched effects are Cohen's d computed within scene category, with bootstrap 95% CIs (2000-3000 resamples) per cell and for the sample-weighted aggregate. ϕ_{typ} is reported as a percentile of the per-scene control Mahalanobis distribution in place of the raw value, which inflates for off-distribution points in high dimensions under shrinkage.

S5 Computational-observer probes (CLIP)

For each affective axis (eerie, nostalgic, abandoned, dreamlike) we built a positive and a negative pole from five paraphrase prompts each, embedded with CLIP's text encoder, and took each pole as the mean unit text embedding (prompt ensembling). For an image embedding v the score is $P(\text{positive pole}) = \sigma\big(\tau(v{\cdot}t_+ - v{\cdot}t_-)\big)$ with τ the CLIP logit scale. Example eerie prompts: positive, "an eerie, unsettling place", "a creepy liminal space", "an uncanny, wrong-feeling scene"; negative, "an ordinary, comfortable place", "a normal everyday scene", "a reassuring lively place". Cross-method agreement (CLIP eerie vs. structural HUMAN_ABSENCE) was Pearson $r = 0.29$ at the image level; the four affect axes alone classified the five clusters at 0.45 (chance 0.20) under the same 5-fold protocol.

S6 Collective human labels (Aesthetics Wiki and are.na)

Sources: the Aesthetics Wiki (`aesthetics.fandom.com`) is a community-maintained encyclopedia whose pages carry a structured {{Aesthetic}} infobox with fields KEY_MOTIFS, KEY_COLOURS, KEY_VALUES, and RELATED_AESTHETICS. We retrieved these via the keyless MediaWiki API for the 26 named aesthetics in our taxonomy, resolving redirects to canonical titles; 21 pages populate KEY_VALUES. For the ecosystem network (main text), we additionally enumerated every main-namespace article via `allpages` and kept the 1,128 carrying an aesthetic infobox (912 with KEY_VALUES), parsing 20,925 RELATED_AESTHETICS entries; restricting to edges whose both endpoints are themselves aesthetics gives the 1,115-node, 5,889-edge graph analysed with Louvain community detection (`networkx`, seed fixed) and a degree-preserving (double-edge-swap) null. are.na (`are.na`) is a curation platform where users file images into named channels; via its keyless v2 API we collected image membership from the top channels matching each aesthetic name ($\approx$2,000 image-label pairs across the four clusters), used here as crowd-assigned category labels.

Affect lexicon: each KEY_VALUES string was scored against a pre-specified lexicon mapping surface terms to four dimensions: unease (uncanny, dread, disorientation, alienation, ...), nostalgia (nostalgia, anemoia, childhood, the past, ...), comfort (simplicity, harmony, optimism, cozy, ...), and refinement (intellectualism, elegance, opulence, tradition, ...). Scores are within-aesthetic proportions; cluster means

and the inverted-vs-rest unease effect ($d = 2.66$) are reported in the main text. The lexicon is fixed in advance and disclosed in full in the code; the graph test below is lexicon-independent.

Related-aesthetic graph: RELATED_AESTHETICS entries were parsed (stripping wiki markup,
, and external links) into a directed similarity graph. Of the resulting edges, 73 have both endpoints inside our 26-aesthetic taxonomy. The within-cluster edge fraction (0.49) was tested against a null that permutes mechanism-cluster labels over aesthetics (5000 shuffles, fixed seed), giving $p = 0.0006$. Image-level membership prediction: the are.na images were downloaded (perceptual-hash de-duplicated, ≥384 px) and passed through the same feature pipeline as the main corpus; the canonical reference for the ϕ coordinates was the original control set (firewall preserved). A random forest (400 trees) predicted the human label under GROUPKFOLD cross-validation with the are.na channel as the group, so every image from a given channel falls in the same fold, a conservative test that prevents the model from memorising a curator's style or near-duplicates ($n = 1{,}886$ images, 52 channel groups). The permutation null shuffles labels within the same grouped protocol (200 shuffles); we report accuracy, the majority-class baseline, the null mean±SD, and z. Both the four-way (mechanism) and 20-way (aesthetic) tasks were run with and without the ϕ coordinates.

Caveats: Aesthetics-Wiki fields are editor-authored summaries with small underlying n; are.na curators are a design-literate subpopulation; channel membership is noisy (no relevance filtering was applied); and the are.na images are copyright-held and used for feature extraction only and are not redistributed. The labels are therefore treated as a corroborating external criterion, with ground-truth affect held out of reach.

S7 Image-feature structure on the licensed corpus

The remainder of the paper grounds this network-level picture in the images themselves. The design matrix is 681×37 (13 colour, 10 segmentation, 4 object, 4 ϕ, plus provenance). ϕ_{typ} and EXP_ABSENT are defined for the scene-tagged inverted/control images; the other clusters carry a generic scene label.

We separate two kinds of claim. Cross-cluster comparisons (restorative vs. hauntological, etc.) are descriptive and carry a confound we name below: cluster was assigned at collection time, so feature differences partly reflect what was collected into each bin. The paper's evidentiary weight rests on the scene-matched inverted-vs-control contrast, where scene type is held fixed and only population varies.

Cross-cluster structure is classifiable with heavy overlap (descriptive): a five-way classifier on the 29 features defined for all images is above chance (cross-validated accuracy 0.73 random forest, 0.68 logistic, vs. 0.20 chance; permutation null 0.24 ± 0.02 over 200 shuffles, $p = 0.005$); the silhouette coefficient is ≈ 0.00, so the clusters are classifiable while heavily overlapping, consistent with the diffuse PCA scatter (Fig. S1; PC1+PC2 = 31.3% variance). The confusion matrix (Table S1) shows the control cluster perfectly recovered and the nostalgia sub-types blurring into one another, with anemoia least separable (24/84, dispersing into restorative and inverted).

The control boundary is largely a firewall artifact. That perfect control recall must be read against limitation (iii): three of the features (ϕ_{surp}, ϕ_{ent}, and, weakly, ϕ_{real}) are near-zero for controls by construction, so the classifier can isolate controls just by reading the firewall. We therefore decompose it. On the 26 non-ϕ features (13 colour, 10 segmentation, 3 object; EXP_ABSENT is scene-tagged only and excluded) control recall falls to 0.40 and overall accuracy to 0.57 (still above chance, permutation $p < 0.01$); on the three ϕ features alone, control recall is 1.00. So the crisp present-day/ aesthetic boundary is mostly tautological, an artifact of the firewall, and we do not claim it as a finding. The robust claim is the above-chance separability of the perceptual (ϕ-free) features (0.57), plus the scene-matched contrast below. Relatedly, ϕ_{real} (realism variance) is essentially flat across clusters (0.032-0.043, overlapping SDs) and we report no effect on it: an all-photographic corpus has little within-image realism inconsistency to detect, which is itself a direct symptom of limitation (i): the post-processing that would create such inconsistency is exactly what the corpus lacks.

Table S1: Five-way cross-validated confusion matrix (random forest; rows true, columns predicted). Control's perfect recovery is largely a firewall artifact (see text: ϕ-free control recall = 0.40); inverted is well recovered; anemoia is the least separable.

true \ pred	restor.	haunt.	invert.	anem.	control
restorative	120	20	15	5	0
hauntological	21	75	22	5	0
inverted	17	8	163	4	2
anemoia	29	11	20	24	0
control	0	0	0	0	120

Table S2: Per-cluster means of load-bearing coordinates (N = 681). ϕ_{typ} and EXP_ABSENT are defined only for scene-tagged inverted/control images (blank otherwise, by design).

cluster	colourful.	emptiness	human_abs.	exp_absent	ϕ_{typ}	ϕ_{ent}
restorative	46.7	0.13	0.12			1.42
hauntological	33.5	0.37	0.37			1.05
inverted	31.6	0.58	0.57	0.74	−1480	0.75
anemoia	35.9	0.23	0.23			1.37
control	31.9	0.34	0.30	0.78	−41	0.02

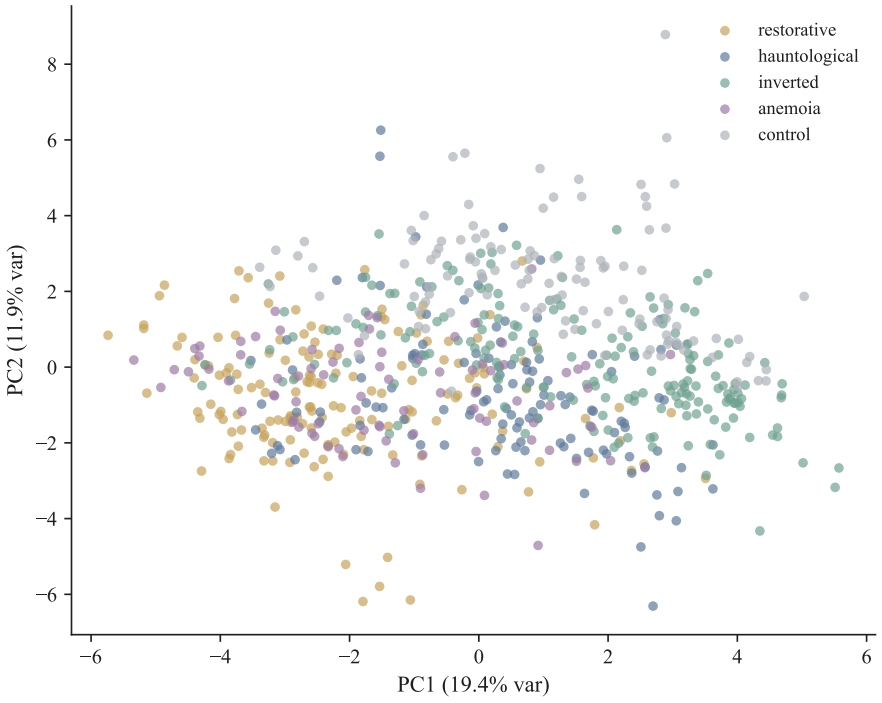


Figure S1: PCA of the standardized 681 × 37 design matrix, colored by cluster. Restorative (gold) partially separates from the desaturated hauntological/anemoia clusters and inverted (green, liminal) leans toward the absence-dominated region, with heavy overlap (silhouette ≈ 0; classifiable, short of a clean partition).

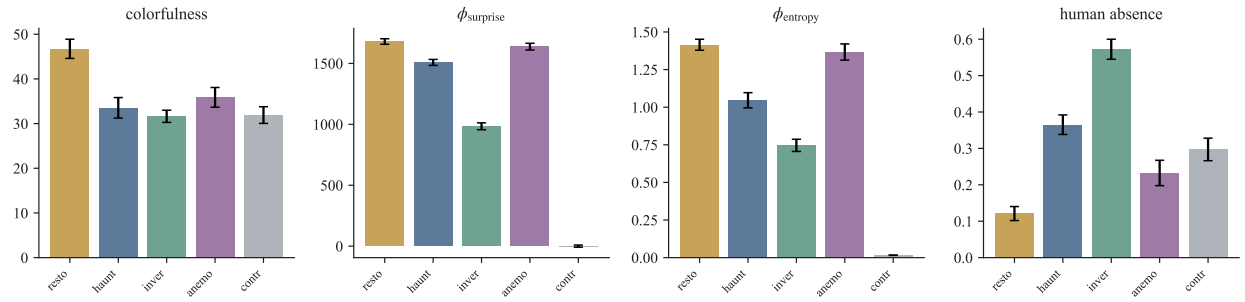


Figure S2: Affective spectrum by cluster (mean ± SEM): colour warmth (restorative high), $\phi_{surprise}$ and $\phi_{entropy}$ (controls ≈ 0 by construction, see firewall discussion), and structural human absence (inverted high). The clusters trace a warmth-to-absence gradient; cross-cluster differences are descriptive (see text).

The per-cluster means (Table S2, Fig. S2) trace a descriptive warmth-to-absence gradient: colour warmth is highest in restorative (colourfulness 46.7 vs. ~32-36 elsewhere) and structural HUMAN_ABSENCE runs the opposite way, peaking in inverted (0.57). We flag this as descriptive and partly confounded: colourfulness is precisely a dimension the aesthetics manipulate in post-processing and our encyclopedic corpus lacks, so the restorative warmth here is as much "photos of nature" as "Frutiger Aero" (see Limitations).

The scene-matched, stratified contrast that controls for subject matter holds scene category fixed and varies only population. Of the 194 inverted images, 170 carry one of seven shared scene categories; the remaining 24 are generically tagged (waiting room, hotel corridor, analog-horror frames) with no populated control counterpart and are excluded from this analysis. The matched cells appear in Table S3. Pooled across categories the effect is large (HUMAN_ABSENCE $d = 0.75$). That pooled estimate is inflated by differing category composition; the within-category effect is a sample-weighted $d = 0.38$ (95% CI $[0.14, 0.68]$) for HUMAN_ABSENCE and 0.29 ($[0.03, 0.59]$) for EMPTINESS: modest, positive, and reliably above zero, with the effect corridor-weighted (corridor is 77/170 of the tagged inverted images, and its own cell $d = 0.35$ has a CI spanning zero).

The structure across cells is the most interesting result. The effect is strong precisely where emptiness violates an occupancy expectation (classroom 0.76, office 0.67, swimming pool 0.56) and vanishes where "empty" is the ordinary state of the scene (parking lot 0.04, playground −0.06). Individual cells are noisy (small n; CIs in Table S3), so this gradient is preliminary, and it is exactly what Fisher's eerie predicts at the scene level: the unease tracks expected presence above emptiness per se. It yields the paper's sharpest pre-registerable hypothesis: human unease ratings should follow the same ordering, highest for empty-when-usually-occupied scenes (§Discussion).

In embedding space the inverted/control gap is larger, with a caveat: ϕ_{typ} is also a distance-from-control measure (squared Mahalanobis to the per-scene control centroid), so it shares the domain-shift concern of limitation (iii), only partly mitigated by being computed within scene. Reported as a percentile to avoid the dimensionality-inflated raw values (−1480 vs. −41): control median 38 (90th percentile 64) vs. inverted median 1387.

The firewall bounds what this can show: controls score $\phi_{surp} \approx 0$ and entropy 0.02 by construction (the PCA basis and scene classifier are fit on controls). The corollary, stated plainly: "all aesthetic clusters are off-manifold" is partly tautological: any image distributionally unlike the controls reads as deviant, and ϕ_{surp} conflates affective deviance with mundane distributional shift. Because every image (controls included) comes from one source (Wikimedia), photographer/scan-artifact shift is reduced while content shift remains; we therefore treat ϕ_{surp} and ϕ_{ent} as hypothesis-generating measures only.

Retro-futurism is conceptually central and empirically under-sampled: the optimistic strand of retro-futurism (Googie, atomic-age, synthwave) collected adequately ($n = 15$, folded into restorative), while

Table S3: Scene-matched inverted-vs-control cells, within-category Cohen's d for HUMAN_ABSENCE with bootstrap 95% CIs (d for EMPTINESS shown for reference). The effect is large where emptiness violates occupancy expectation and absent where empty is ordinary; small cells are individually unstable. Pooled $d = 0.75$ is composition-inflated.

scene	inv. n	ctl. n	d human_abs. [95% CI]	d empt.
classroom	12	18	0.76 [0.08, 1.80] †	0.36
office	11	27	0.67 [0.06, 1.57] †	0.33
swimming pool	19	7	0.56 [0.01, 1.07] †	0.52
shopping mall	24	19	0.37 [−0.23, 1.01]	0.00
corridor	77	18	0.35 [−0.12, 0.94]	0.41
parking lot	19	15	0.04 [−0.62, 0.67]	0.17
playground	8	16	−0.06 [−0.86, 0.84] †	0.13
weighted within-category			0.38 [0.14, 0.68]	0.29 [0.03, 0.59]

† small cell (min $n < 15$); per-cell d unstable, interpret via CI.

the decayed/hauntological strand (cassette futurism, dieselpunk) is almost absent from open-license banks ($n = 1$). We therefore do not report the within-domain restorative-vs-hauntological retro-futurism contrast here: the corpus cannot yet support it. The same-subject, opposite-affect comparison would make a clean future test, and a targeted, properly licensed collection is a priority for follow-up work.

The two measures of the eerie dissociate: segmentation HUMAN_ABSENCE separates inverted from control (0.57 vs. 0.30), while the object-level EXP_ABSENT stays flat (0.74 vs. 0.78), since openly licensed "populated" scenes are often architectural photographs with few people, so the detector finds little to violate. Structural/configural absence is the cleaner measure of the eerie in this corpus.

To test whether the affective structure is an artifact of our hand-built features or a property of the stimuli recoverable by an independent mechanism, we probed every image with CLIP [21] as a zero-shot observer: for each of four theory-relevant axes (eerie, nostalgic, abandoned, dreamlike) we computed $P(\text{positive pole})$ over prompt-ensembled contrastive text probes. CLIP is trained on web-scale human image-text pairs, so its alignments encode aggregated human associations, a semantics-based signal computed along an entirely different path from the perceptual axes and ϕ coordinates. It stands in for human affect at one remove.

The main contrast replicates: CLIP rates the inverted cluster most eerie (0.67) and restorative least (0.44), with an inverted-vs-control eerie difference of $d = 0.72$ (95% CI $[0.47, 0.99]$), comparable to or larger than the structural metric, by a different measurement mechanism. CLIP nostalgia peaks for the hauntological cluster (vaporwave), as expected, and the four zero-shot affect axes alone classify the five clusters at 0.45 (chance 0.20). Two independent methods thus recover the same cluster ordering.

The two methods are complementary, and they disagree where it matters. At the image level CLIP eeriness correlates only modestly with structural HUMAN_ABSENCE ($r = 0.29$): the methods triangulate one construct while measuring distinct features. And CLIP eeriness does not reproduce the scene-level occupancy gradient: the within-scene inverted-vs-control effect is present, with its ordering across scenes differing (e.g. corridor near-zero, playground large), opposite the structural pattern. Structural absence and semantic eeriness agree that the inverted pole is eerie and differ on where, indicating that the occupancy moderator is method-dependent. Two independent measurements thus converge on the same inverted-vs-rest contrast.

S8 Frontier-panel confusion matrix

Figure S4 gives the full three-model confusion behind the flow in Fig. 3: human-assigned mode against VLM consensus mode, row-normalised, with the per-mode mode-match recall.

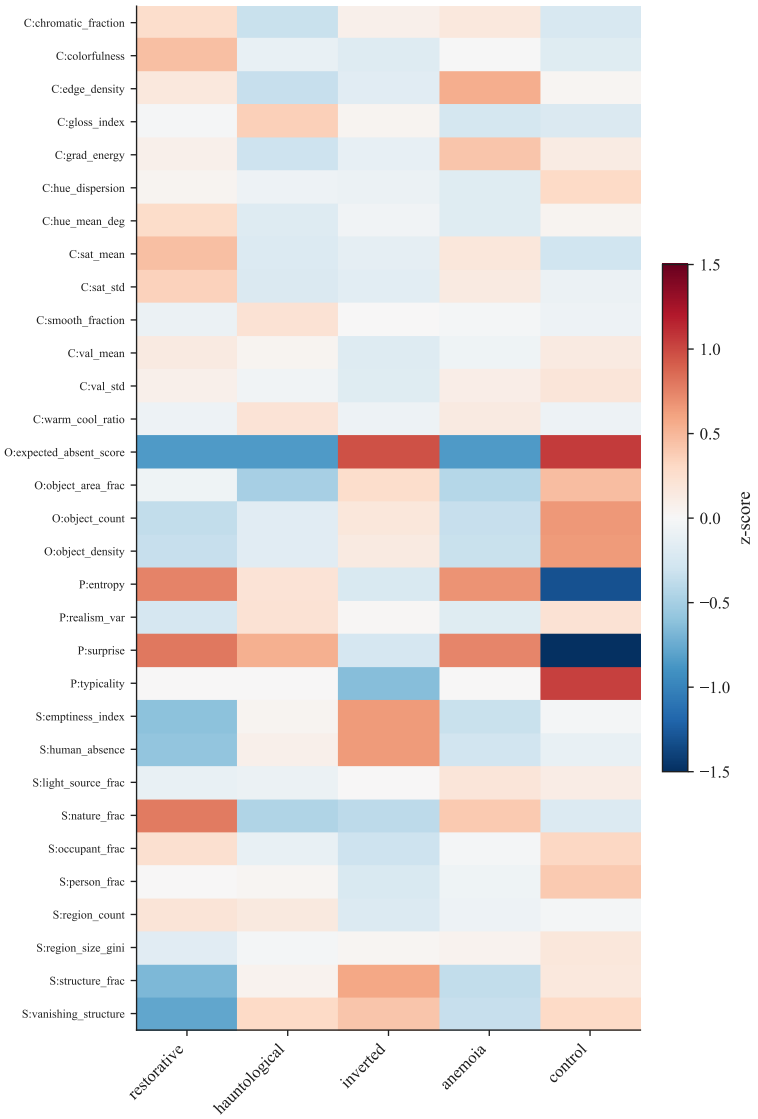


Figure S3: Standardized (z-scored) feature means by cluster across all 31 numeric features (C: colour, S: segmentation, O: object, P: ϕ). Restorative is marked by colour warmth, inverted by structural absence and strongly negative typicality.

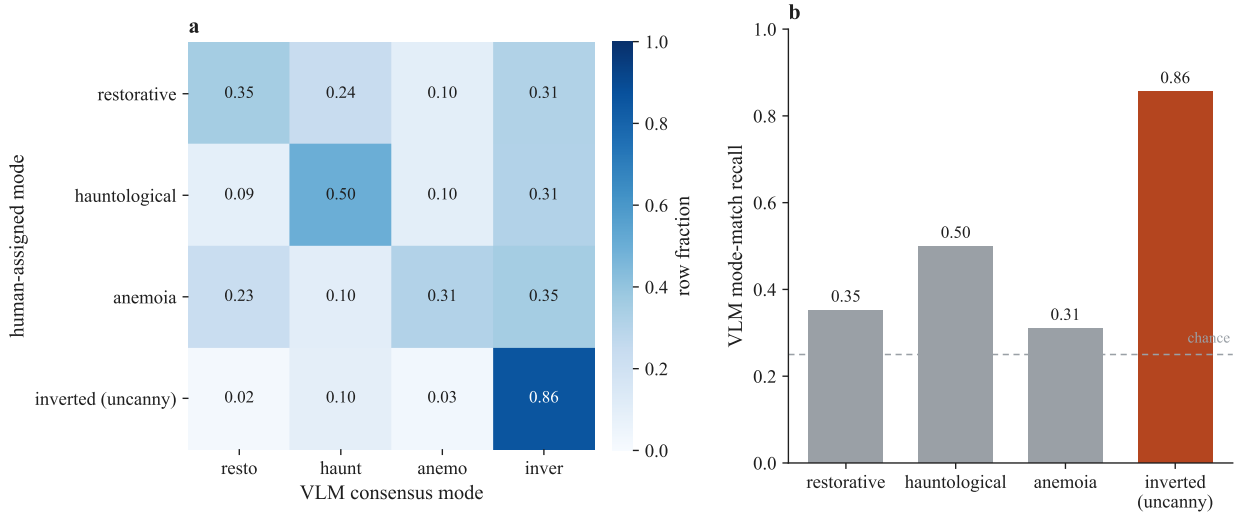


Figure S4: Three-model VLM confusion on the are.na images. (a) Human-assigned mode (rows) vs. VLM consensus mode (columns), row-normalised; every non-uncanny class leaks into the "inverted" column. (b) Per-mode mode-match recall. This is the matrix that Fig. 3 summarises as a flow.